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Engineering Talent & Community Team
AI & Frontier Models Community
G2i Inc.
Miami, FL, USA (Remote)

Dear G2i Talent and Engineering Team,

G2i's platform plays a pivotal role in equipping top AI labs with vetted engineers capable of building verifiable evaluation harnesses, RL environments, and high-fidelity post-training datasets for frontier models. As frontier model capabilities evolve toward complex multi-turn reasoning and tool use, the bottleneck has shifted from raw data volume to rigorous post-training environments with deterministic evaluation harnesses. Having engineered open-source retrieval evaluation benchmarks and autonomous agent systems, I have directly addressed the technical challenges of paired statistical validation, tool-use execution sandboxing, and latency profiling that frontier labs require.

In my open-source benchmark **ncert-rag-evals**, I constructed an empirical evaluation testbed across 17 STEM textbooks in **Python** and **Jupyter Notebooks**, implementing custom hybrid search in **SQLite FTS5** and **ChromaDB** to evaluate 8 retrieval strategies on raw CPU hardware. By fusing BM25 lexical search with dense embeddings via Reciprocal Rank Fusion (RRF), I proved an empirical **86.9% Recall@5 in 8.9ms on CPU**, matching LLM query expansion while cutting latency by **99.8%**. Additionally, in **vectorless-rag**, I benchmarked dense vector search against LLM-based page indexing using **Pydantic AI** and **Ollama**, mapping the **2.7x token cost** and **2x latency** trade-offs across benchmark queries while streaming page windows via **PyMuPDF** with bounded memory (**under 200MB**).

Frontier training and evaluation workflows demand deterministic verification over ungrounded vanity metrics. To ensure empirical defensibility in **ncert-rag-evals**, I validated retrieval significance using paired McNemar statistical hypothesis tests (**p = 0.0021**) supported by 92 unit tests in **Pytest**. Furthermore, during my software engineering internship at **LexoLegal**, I built autonomous multi-turn assistants using **Pydantic AI** and **FastAPI**, isolating verified user identities inside AgentDeps to eliminate prompt injection risks and orchestrating **Model Context Protocol (MCP)** clients alongside **Celery** and **Redis** distributed locks to handle burst traffic with **sub-2s latency**.

I am excited to bring this evaluation rigor, agentic architecture background, and benchmark discipline to the G2i Talent Pool for client projects across top AI labs. As an engineer based in India, I operate smoothly across asynchronous remote workflows, and I look forward to completing the technical assessment and contributing to frontier AI evaluation and training initiatives.

Sincerely,

Aman Verma